Pre-calculus 2

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Lecture 12: The Laws of Sin and Cos

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The triangle in Figure 1 is a non-right triangle since none of its angles measure 90°. We'll start by deriving the **Laws of Sin and Cos** so that we can study non-right triangles.

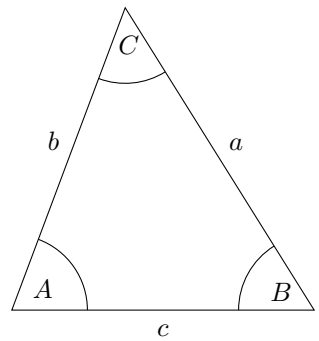

Figure 1

The Law of Sin

To derive the **Law of Sin**, let's construct a segment h in the triangle, which connects the vertex of angle C to the side c. This segment should be perpendicular to side c and is called a *height* of the triangle. See Figure 2.

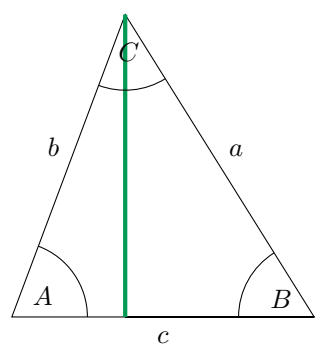

Figure 2

The segment h splits the triangle into two right triangles on which we can apply what we know about right triangle trigonometry. See Figure 3

We can use the 2 right triangles to obtain expressions for both $\sin(A)$ and $\sin(B)$.

$$\sin(A) = \frac{h}{b} \text{ and } \sin(B) = \frac{h}{a}$$

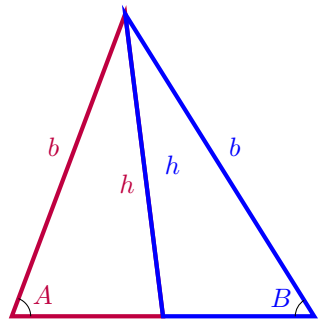

Figure 3

. We can now solve both of these equations for h:

$$\begin{aligned} & \sin(A) = \frac{h}{b} \text{ and } \sin(B) = \frac{h}{a} \\ \Rightarrow \quad & h = b\sin(A) \text{ and } h = a\sin(B). \end{aligned}$$

Now, since both of the h's represent the length of the same segment, they are equal. By setting h's equal to each other we obtain the following:

$$b\sin(a) = a\sin(B)$$

. This equation provides us with what is known as the **Law of Sin**. Typically, the law is written in terms of ratios. If we divide both sides by $a \times b$, we obtain the following:

$$\begin{aligned} & b\sin(A) = a\sin(B) \\ \Rightarrow \quad & \frac{b\sin(A)}{a \times b} = \frac{a\sin(B)}{a \times b} \\ \Rightarrow \quad & \frac{\sin(A)}{a} = \frac{\sin(B)}{b}. \end{aligned}$$

Definition 1 (Law of Sines)**.** If a triangle's sides and angles are labeled like the triangle in Figure 4, then:

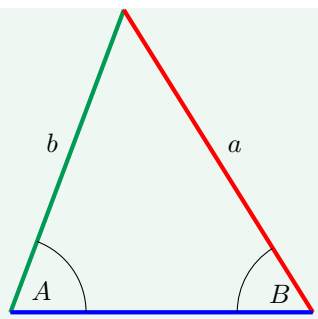

Figure 4

$$\frac{\sin(A)}{a} = \frac{\sin(B)}{b}$$

. This is an **identity** since it is true for *all* triangles.

The Law of Cos

To derive the **Law of Cos**, let's start with a generic triangle and draw the height, h, just as we did when we derived the Law of Sin. See Figure 5.

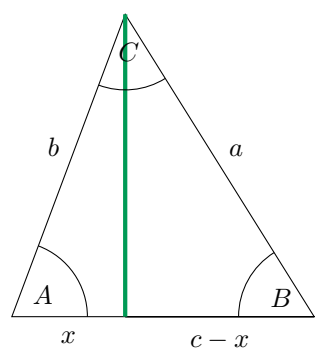

Figure 5

Again we want to consider the two right triangles induced by constructing the line h on the triangle given in Figure 5. This time, we want to use the two pieces that the side c is split into. Let's call the segment on the left x and then the segments on the right must be $c - x$ units long. We've emphasized the two right triangles and labeled the two pieces of side c in Figure 6

First, notice that the following is true:

$$\begin{aligned} \cos(A) &= \frac{x}{b} \\ \Rightarrow \qquad x &= b\cos(A). \end{aligned}$$

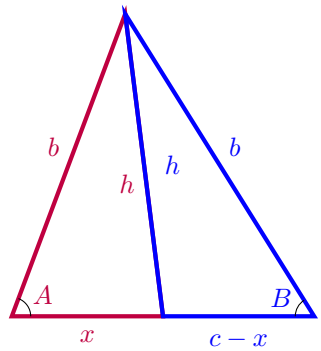

Figure 6

Now, let's apply the Pythagorean Theorem to each of the two triangles in Figure 6. The blue triangle on the left gives us:

$$x^2 + h^2 = b^2$$

. and we can solve this for h^2 and obtain:

$$h^2 = b^2 - x^2$$

. The blue triangle on the right gives us:

$$(c - x)^2 + h^2 = a^2$$

. and we can use the fact that $h^2 = b^2 - x^2$ to eliminate h from this equation:

$$\begin{aligned} & (c - x)^2 + h^2 = a^2 \\ \Rightarrow \quad & (c - x)^2 + (b^2 - x^2) = a^2. \end{aligned}$$

Finally, we can simplify the left side of this equation and use the fact that $x = b\cos(A)$ to eliminate x:

$$\begin{aligned} & (c - x)^2 + (b^2 - x^2) = a^2 \\ \Rightarrow \quad & c^2 - 2cx + x^2 + b^2 - x^2 = a^2 \\ \Rightarrow \quad & c^2 - 2cx + b^2 = a^2 \\ \Rightarrow \quad & c^2 - 3c \times b\cos(A) + b^2 = a^2 \end{aligned}$$

.

This last equation is known as the **Law of Cosines**.

Definition 2 (Law of Cosines)**.** If a triangle's sides and angles are labeled like the triangle in Figure 7, then:

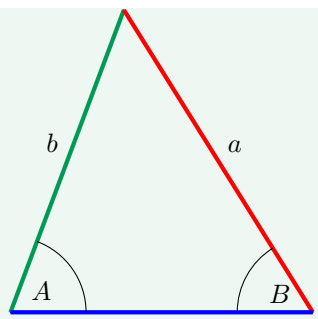

Figure 7

$$a^2 = b^2 + c^2 - 2bc \times \cos(A)$$

. This is an **identity**, since it is true for *all* triangles.

Solutions

Solution 1 (Exercise ??).

1. This represents a set since it's well defined. We all know what it means to be registered for a class.
2. This does not represent a set since it's not well defined. There are many different interpretations of what it means to be a good student (get an **A** or pass the class or attend class or avoid falling asleep in class or don't cause trouble in class)

Solution 2 (Exercise ??).

- $(-4, \infty) \cup [-8, 3] = [-8, \infty)$
- $(-4, \infty) \cup (-\infty, 2] = (-\infty, \infty) = \mathbb{R}$
- $(-4, \infty) \cap (-\infty, 2] = (-4, 2]$
- $(-4, \infty) \cap [-10, -5] = \emptyset$

Solution 3 (Exercise ??). Since $2\pi = 360°$ $\frac{2\pi}{360°} = 1 = \frac{360°}{2\pi}$. We're trying to cancel out the radians. So, to do this, we will need to have the radians on the bottom, and the degrees on the top. After we do the multiplication, we will be left with the degrees.

$$\begin{aligned} 8\,\text{rad} \times \left(\frac{360°}{2\pi\,\text{rad}}\right) &= \frac{8 \times 360°}{2\pi} \\ &= \frac{1440°}{\pi} \\ &\approx 458.4. \end{aligned}$$

Now, let's convert 8° into 8 radians.

$$\begin{aligned} 8° \left(\frac{\pi\,\text{rad}}{180°}\right) &= \frac{8\pi}{180}\,\text{rad} \\ &= \frac{\pi}{45}\,\text{rad} \end{aligned}$$

.

Solution 4 (Exercise ??). Convert 40° into radians:

$$\begin{aligned} 40^\circ \times \left(\frac{\pi}{180^\circ}\right) &= \frac{40\pi}{180}\,\text{rad} \\ &= \frac{4\pi}{18} \\ &= \frac{2\pi}{9} \\ &\approx 0.7 \end{aligned}$$

.

Now we're ready to compute the arc length:

$$\begin{aligned} s &= r\times \mid \theta \mid \\ &= (30m) \times (\frac{2\pi}{9}) \\ &= \frac{20\pi}{3}m. \end{aligned}$$

Solution 5 (Exercise ??). The **period** for the graph is 2. The reason is because if you shift the graph 2 units either left or right, you will get the same graph.

Another way to find the period is the take an x-intercept and the x-intercept to the right of it and subtract the two: $10 - 8 = 2$

Solution 6 (Exercise ??). The **midline** of the graph is 0.

$$\frac{1 + (-1)}{2} = \frac{0}{2} = 0$$

.

The **amplitude** of the graph is 1, which is the distance from the maximum or minimum to the midline.

Solution 7 (Exercise ??).

$$\begin{aligned} P &= (\cos(\theta), \sin(\theta)) \\ &= \cos(\theta) = -\frac{3}{5} \\ &= \sin(\theta) = \frac{4}{5}. \end{aligned}$$

Solution 8 (Exercise ??). The point Q is specified by α on the circumference of a circle of radius 6 units. Thus ...

$$Q = (6\cos(\alpha), 6\sin(\alpha))$$

.

Solution 9 (Exercise ??). Since the Pythagorean Identity gives us an equation involving sin and cos, we can use it to find one of the values when we know the other value. In this case, we know the value of $\sin(A)$, so we can use the Pythagorean Identity to find $\cos(A)$:

$$\begin{aligned}
&\sin^2(A) + \cos^2(A) = 1 \\
\Rightarrow &\left(\frac{1}{2}\right)^2 + \cos^2(A) = 1 \\
\Rightarrow &\cos^2(A) = 1 - \left(\frac{1}{3}\right)^2 \\
\Rightarrow &\cos(A) = -\sqrt{\frac{8}{9}} \\
\Rightarrow &\cos(A) = -\frac{2\sqrt{2}}{3}.
\end{aligned}$$

The reason I made the answer negative is because the value of cos in Quadrant II is negative. If cos was in Quadrant I, then the answer would be positive.

Solution 10 (Exercise ??).

$$\begin{aligned}
\tan\left(\frac{\pi}{6}\right) &= \frac{\sin\left(\frac{\pi}{6}\right)}{\cos\left(\frac{\pi}{6}\right)} \\
&= \frac{\frac{1}{2}}{\frac{\sqrt{3}}{2}} \\
&= \frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3} \\
\sec\left(\frac{\pi}{6}\right) &= \frac{1}{\cos\left(\frac{\pi}{6}\right)} \\
&= \frac{1}{\frac{\sqrt{3}}{2}} \\
&= \frac{2}{\sqrt{3}} = \frac{2\sqrt{3}}{3} \\
\csc\left(\frac{\pi}{6}\right) &= \frac{1}{\sin\left(\frac{\pi}{6}\right)} \\
&= \frac{1}{\frac{1}{2}} \\
&= 2 \\
\cot\left(\frac{\pi}{6}\right) &= \frac{\cos\left(\frac{\pi}{6}\right)}{\sin\left(\frac{\pi}{6}\right)} \\
&= \frac{\frac{\sqrt{3}}{2}}{\frac{1}{2}} \\
&= \sqrt{3}
\end{aligned}$$

.

Solution 11 (Exercise ??).

$$\begin{aligned}
\tan\left(\frac{\pi}{4}\right) &= \frac{\sin\left(\frac{\pi}{4}\right)}{\cos\left(\frac{\pi}{4}\right)} \\
&= \frac{\frac{\sqrt{2}}{2}}{\frac{\sqrt{2}}{2}} \\
&= 1 \\
\sec\left(\frac{\pi}{4}\right) &= \frac{1}{\cos\left(\frac{\pi}{4}\right)} \\
&= \frac{1}{\frac{\sqrt{2}}{2}} \\
&= \frac{2}{\sqrt{2}} = \sqrt{2} \\
\csc\left(\frac{\pi}{4}\right) &= \frac{1}{\sin\left(\frac{\pi}{4}\right)} \\
&= \frac{1}{\frac{\sqrt{2}}{2}} \\
&= \frac{2}{\sqrt{2}} = \sqrt{2} \\
\cot\left(\frac{\pi}{4}\right) &= \frac{\cos\left(\frac{\pi}{4}\right)}{\sin\left(\frac{\pi}{4}\right)} \\
&= \frac{\frac{\sqrt{2}}{2}}{\frac{\sqrt{2}}{2}} \\
&= 1
\end{aligned}$$

.

Solution 12 (Exercise ??).

$$\begin{aligned}
\tan\left(\frac{\pi}{3}\right) &= \frac{\sin\left(\frac{\pi}{3}\right)}{\cos\left(\frac{\pi}{3}\right)} \\
&= \frac{\frac{\sqrt{3}}{2}}{\frac{1}{2}} \\
&= \sqrt{3} \\
\sec\left(\frac{\pi}{3}\right) &= \frac{1}{\cos\left(\frac{\pi}{3}\right)} \\
&= \frac{1}{\frac{1}{2}} \\
&= 2 \\
\csc\left(\frac{\pi}{3}\right) &= \frac{1}{\sin\left(\frac{\pi}{3}\right)} \\
&= \frac{1}{\frac{\sqrt{3}}{2}} \\
&= \frac{2}{\sqrt{3}} = \frac{2\sqrt{3}}{3}
\end{aligned}$$

.

Solution 13 (Exercise ??). The reference angle for $150°$ is $30°$.
The reference angle for $\frac{5\pi}{4}$ is $\frac{\pi}{4}$.

Solution 14 (Exercise ??). It should be clear that function g is a sinusoidal function of the form $y = A\sin(\omega(t-h)) + k$ where $A = 2, \omega = 1, h = 0, k = -3$.

After inspecting the rules for the functions f and g, we should notice that we could construct the function $g(t) = 2\sin(t) - 3$ by multiplying the outputs of the function $f(t) = \sin(t)$ by 2 and then subtracting 3 from the result. We can express this algebraically with the equation below:

$$g(t) = 2f(t) - 3$$

.

Based on what we know about the graph transformations, we can conclude that we can obtain graph of g by starting with the graph of f and first stretching it vertically by a factor of 2 and then shifting it down 3 units. Since $f(t) = \sin(t)$ has amplitude of 1 unit, if we stretch it vertically by a factor of 2, then we'll double the amplitude, so we should expect that the

amplitude of g to be 2 units. Also, since $f(t) = \sin(t)$ has midline $y = 0$, when we shift it down 3 units, the resulting midline for the function g will be $y = -3$.

period: 2π units
midline: $y = -3$
amplitude: 2 units
horizontal shift: 0 units

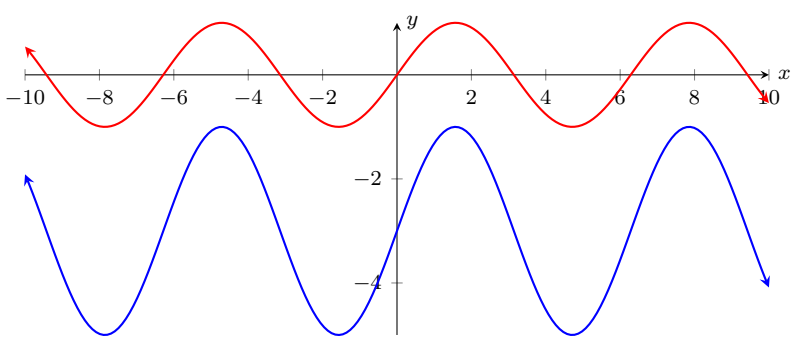

Figure 8: The graph of $f(t) = \sin(t)$ and $g(t) = 2\sin(t) - 3$

Solution 15 (Exercise ??). Let's start by finding the values of A, ω, k. To do this, we need to find the midline, amplitude, and period.

The midline is $y = -3$, which dictates what the value of k is.
The amplitude is the distance from the maximum/minimum to the midline. The amplitude is 4 units.
The period is the distance from each maximum/minimum. $6.5 - 0.5 = 6$ units. The period doesn't actually appear as a number in the algebraic rule for the function. But the period is connected with the value that we use of ω. ω causes a horizontal stretch which gives us the desired period.

$$6 = 2\pi \times \frac{1}{\omega} = \frac{2\pi}{6} = \boxed{\frac{\pi}{3}}$$

. When we're trying to find a formula that involves sin, we want to think in terms of sin waves, which start at the midline and come up out of the midline. So, we'll want to find a place in this function where the function is crossing through the midline and coming up out of the midline. That would be at the point: $(0, -1)$. So, for the sin function, $h = -1$. For cos, it's the opposite. Cos starts at the maximum, and goes down to the midline. That

would be at the point: $(0, 0.5)$. So, for the cos function, $h = 0.5$.

$$g(t) = 4\sin\left(\frac{\pi}{3}(t+1)\right) - 3 \qquad g(t) = 4\cos\left(\frac{\pi}{3}\left(t - \frac{1}{2}\right)\right) - 3$$

.

Solution 16 (Exercise ??).

- To evaluate $\sin^{-1}(-\frac{1}{2})$, we need to find a value, p, such that $-\frac{\pi}{2} \leq p \leq \frac{\pi}{2}$ and $\sin(p) = -\frac{1}{2}$. $p = -\frac{\pi}{6}$. Thus, $\sin^{-1}(-\frac{1}{2}) = -\frac{\pi}{6}$.
- To evaluate $\sin(\sin^{-1}(\frac{\sqrt{3}}{2}))$, we need to first find $\sin^{-1}(\frac{\sqrt{3}}{2})$, such that $-\frac{\pi}{2} \leq p \leq \frac{\pi}{2}$. $p = \frac{\pi}{3}$. Thus, $\sin^{-1}(\frac{\sqrt{3}}{2}) = \frac{\pi}{3}$. Now we can evaluate that:
$$\sin\left(\sin^{-1}\left(\frac{\sqrt{3}}{2}\right)\right) = \sin\left(\left(\frac{\pi}{3}\right)\right) = \frac{\sqrt{3}}{2}$$
.

Solution 17 (Exercise ??).

- To evaluate $\cos^{-1}(0)$, we need to find a value, p, such that $0 \leq p \leq \pi$ and $\cos(p) = 0$. $p = \frac{\pi}{2}$. Thus, $\cos^{-1}(0) = \frac{\pi}{2}$.
- To evaluate $\cos(\cos^{-1}(\frac{1}{2}))$, we need to first find $\cos^{-1}(\frac{1}{2})$, such that $0 \leq p \leq \pi$. $p = \frac{\pi}{3}$. Thus, $\cos^{-1}(\frac{1}{2}) = \frac{\pi}{3}$. Now we can evaluate that:
$$\cos\left(\cos^{-1}\left(\frac{1}{2}\right)\right) = \cos\left(\frac{\pi}{3}\right) = -\sqrt{3}$$
.

Solution 18 (Exercise ??).

- To evaluate $\tan^{-1}(1)$, we need to find a value, p, such that $-\frac{\pi}{2} \leq p \leq \frac{\pi}{2}$ and $\tan(p) = 1$. $p = \frac{\pi}{4}$. Thus, $\tan^{-1}(1) = \frac{\pi}{4}$.
- To evaluate $\tan(\tan^{-1}(-\sqrt{3}))$, we need to first find $\tan^{-1}(-\sqrt{3})$, such that $-\frac{\pi}{2} \leq p \leq \frac{\pi}{2}$. $p = -\frac{\pi}{3}$. Thus, $\tan^{-1}(-\sqrt{3}) = -\frac{\pi}{3}$. Now we can evaluate that:
$$\tan\left(\tan^{-1}\left(-\sqrt{3}\right)\right) = \tan\left(-\frac{\pi}{3}\right) = -\sqrt{3}$$
.

Solution 19 (Exercise ??). The sin value -0.555 isn't a "friendly" sin value. We don't know what input for the sin function is related to the output -0.555, so we need to utilize the inverse sin function in order to solve the equation:

$$\begin{aligned} \sin(t) &= -0.555 \\ \Rightarrow \sin^{-1}(\sin(t)) &= \sin^{-1}(-0.555) \\ \Rightarrow \qquad t &= \sin^{-1}(-0.555) \approx -0.588 \end{aligned} .$$

Although we've found one solution to the equation, we aren't done yet! The inverse sin inverse only gives us one value, but we know that the periodic nature of the sin function suggests that there are infinitely many solutions to an equation like this.

We can find all of the solutions by using the solution that we found using the inverse sin function along with the fact that the sin function has period 2π. Since the sin function has a period of 2π units, we know that the outputs repeat every 2π units. So if $t \approx -0.588$ is a solution, the values represented by $t \approx -0.55 + 2k\pi, k \in \mathbb{Z}$. That gives us more solutions, but we're still missing half of the solutions. Since $\sin(t) = (\pi - t)$, we can use this identity to find our other half missing solution. $t = \pi - (-0.588) + 2k\pi, k \in \mathbb{Z}$. Therefore, the complete solution to the equation is:

$$t \approx -0.588 + 2k\pi \text{ or } t \approx \pi + 0.588 + 2k\pi \text{ for all } k \in \mathbb{Z}$$

.

Solution 20 (Exercise ??). To find the value for all the 6 trig functions for the angle α, we need to find the value for the hypotenuse. For this, we can use the Pythagorean Theorem:

$$\begin{aligned} & a^2 + b^2 = c^2 \\ \Rightarrow \quad & 12^2 + 9^2 = c^2 \\ \Rightarrow \quad & 144 + 81 = c^2 \\ \Rightarrow \quad & 225 = c^2 \\ \Rightarrow \quad & \sqrt{225} = \sqrt{c^2} \\ \Rightarrow \quad & c = 15 \end{aligned}$$

.

Now we can use **SOH CAH TOA** to find the value for all 6 trig

functions.

$$\begin{aligned}
\sin(\alpha) &= \frac{\text{OPP}}{\text{HYP}} \Rightarrow \sin(\alpha) = \frac{9}{15} = \frac{3}{5} \\
\cos(\alpha) &= \frac{\text{ADJ}}{\text{HYP}} \Rightarrow \cos(\alpha) = \frac{12}{15} = \frac{4}{5} \\
\tan(\alpha) &= \frac{\text{OPP}}{\text{ADJ}} \Rightarrow \tan(\alpha) = \frac{9}{12} = \frac{3}{4} \\
\sec(\alpha) &= \frac{1}{\cos(\alpha)} \Rightarrow \sec(\alpha) = \frac{1}{\frac{4}{5}} = \frac{5}{4} \\
\csc(\alpha) &= \frac{1}{\sin(\alpha)} \Rightarrow \csc(\alpha) = \frac{1}{\frac{3}{5}} = \frac{5}{3} \\
\cot(\alpha) &= \frac{1}{\tan(\alpha)} \Rightarrow \cot(\alpha) = \frac{1}{\frac{3}{4}} = \frac{4}{3}
\end{aligned}$$

.

Solution 21 (Exercise ??). We know that we have an angle with a 90° and a 60°, which implies that $A = 30°$.

We need to create an equation with only either b or c as the unknown:

$$\begin{aligned}
& \sin(30°) = \frac{7}{c} \\
\Rightarrow \quad & \frac{1}{2} = \frac{7}{c} \\
\Rightarrow \quad & c = 14
\end{aligned}$$

.

Now we can use the Pythagorean Theorem to solve for b :

$$\begin{aligned}
& a^2 + b^2 = c^2 \\
\Rightarrow \quad & 7^2 + b^2 = 14^2 \\
\Rightarrow \quad & 49 + b^2 = 196 \\
\Rightarrow \quad & b^2 = 147 \\
\Rightarrow \quad & \sqrt{b^2} = \sqrt{147} \\
\Rightarrow \quad & b = 7\sqrt{3}
\end{aligned}$$

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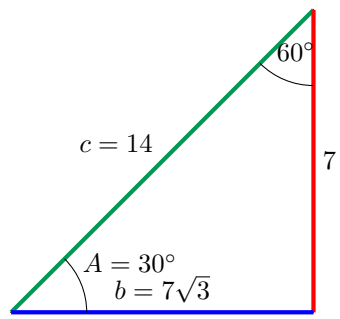

Figure 9